Stage 2 is the automated web crawler that sits between the source directory (Stage 1) and the AI extraction engine (Stage 3). Its job is to visit every grant-making organisation's website on a regular schedule, collect the relevant content, and hand it off cleanly for analysis. Here is what was built, in plain terms.

What it does, at a high level

Every week, the system visits all 582 grant-making organisations in the database. Every day, it revisits the roughly 40 most active ones. For each website, it reads the pages most likely to contain grant opportunities, saves the content, and notes what has changed since the last visit.

How it browses websites

Websites differ enormously in how they present information. Some are straightforward pages; others require a real browser to load their content because they use heavy JavaScript. The crawler handles four types of website behaviour — simple page fetching, full browser rendering, optional rendering, and sites behind bot-protection — and selects the right approach automatically for each organisation. It also behaves like a genuine human visitor rather than a robot: it varies the time between page visits, rotates browser identity signatures, pauses as if reading, and respects each website's own published rules about what may be crawled.

Staying up to date efficiently

Rather than re-reading entire websites from scratch each week, the system uses two shortcuts. First, it checks each organisation's news feed (RSS) to detect new announcements since the last visit, so it can prioritise fresh content. Second, it consults the site's index file (sitemap) to find grant-related pages directly, rather than following every link one by one.

When content is saved, it is fingerprinted with a unique identifier. On the next crawl, if the fingerprint has not changed, the page is skipped — so only genuinely updated content is re-processed, saving time and storage.

Handling documents

Many grant announcements are published as PDF files. The crawler downloads these automatically, extracts the text from them (and uses optical character recognition for scanned documents that contain images rather than typed text), detects the language, and strips out repetitive headers and footers. Documents in French, Spanish, Arabic, Portuguese, Chinese, Japanese, and Korean are flagged for translation before AI analysis.

Security and access

Some websites require a login session to access grant listings. The crawler can securely store and reuse login cookies, encrypted at rest, so it can maintain access across visits without re-authenticating manually each time. For websites protected against automated access, it can route traffic through a residential proxy pool to avoid being blocked.

Organised storage

Everything collected is stored in a consistent folder structure — one folder per organisation, one sub-folder per weekly crawl cycle. HTML pages are kept compressed, PDFs are stored with their extracted text alongside them, and each file is accompanied by a small record noting when it was collected, what language it is in, and whether the content changed. Old copies are automatically deleted: the two most recent HTML snapshots and four most recent PDF copies are retained per organisation; everything older is removed to conserve disk space.

Knowing when something goes wrong

The system monitors itself throughout every crawl. If a single website fails to load — zero pages retrieved — an alert is sent to the operator that same day, by both email and a messaging channel (Slack or Discord). The email subject line tells you immediately whether the problem is critical or merely worth attention, so you can triage from your inbox. A plain-English summary file is also written at the end of every crawl, listing which sites failed, PDF extraction problems, CAPTCHA blocks, and anything else that needs a human eye.

If a website fails consistently across three successive weeks, it is flagged as a candidate for permanent removal from the schedule and added to a manual review list, so you can investigate whether the site has moved or shut down.

Adapting the schedule

The system learns from patterns over time. If an organisation's website has not changed content across several consecutive weeks, it is automatically moved to a less frequent crawl schedule — bi-weekly, then monthly — so resources are concentrated on the sites that actually publish new opportunities regularly. As soon as a site shows new content again, it is returned to the weekly schedule.

Quality assurance

At the close of every crawl, a quality report is produced in two forms: a machine-readable file consumed by Stage 3, and a human-readable plain-text summary for operator review. The report captures how many pages were collected per organisation, what proportion were grant-relevant, PDF extraction success rates, and a complete list of any alerts that were triggered.

In summary, Stage 2 is a fully automated, self-monitoring intelligence-gathering system that keeps the grant database current, handles the full variety of real-world website behaviour, notifies you promptly when anything goes wrong, and manages its own storage and scheduling without requiring ongoing manual intervention.